

Language Model Forge: An Early-Stage Design Science Study to Simplify Language Model Customization

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Agenda



Design Motivation

Existing Designs

Kernel Design Theories

Methodology (Pure Technology Design Study)

Initial Results

Conclusions

General Guidance



Yaman, Cletus, and Sibonginkosi:

Our presentation time is 15 minutes. Plan to leave at least 2-5 minutes at the end for questions.

To keep consistent formatting, let's use the following conventions:

Bullet points with only key details on the slide or tables and graphs. You should never have to reduce font below 18pt. If you have too much info, remove some or create a new slide.

We will remove this slide later

Motivation



Sibonginkosi this will be your slide to introduce why we are building LMForge.

For example, most non-technical academics and business and engineering professionals don't possess data science workflow knowledge. But many AI workflow creation tools require that knowledge. This limits access to custom AI tooling for many academic and business applications.

Look to the paper Introduction section for other ideas.

Be sure to focus on the core of this study, which was automating fine-tuning dataset creation workflows

Existing Designs



Sibonginkosi this will be your slide to introduce research related to AI workflows.

You can bring up some of the existing AI workflow tools like LangFlow, n8n, and the other one that came from academics in the paper.

Look to the introduction and literature review sections of the paper for guidance

Kernel Design Theories



Sibonginkosi this will be your slide to introduce the kernel design theories

Probably start with a statement that design science doesn't necessarily test theories, but uses theories to guide design.

Briefly introduce each theory with the key elements of the theory that guided our design. Check the Kernel Design Theory section of the paper.

Methodology: Two Model Approach



Yaman and Cletus, use this slide to give a high level view of what you were trying to accomplish with the study.

Introduce the two model approach with the encoder to identify relevant source material and the deoder to convert chunks of relevant source material into JSON formatted fine-tuning data, specifically for question-answer fine-tuning.

Methodology: Hardware Configuration



Yaman briefly introduce the hardware on this slide. See the methodology section of the paper.

Methodology: Encoder Study



Cletus, introduce how you ran the study on the encoder model study

Methodology: Decoder Study



Yaman, introduce how you ran the study on the decoder model study

Initial Results: Two Model Approach



Yaman and Cletus, given a high level view of what you learned from the two model approach

You can also mention that we have since revised some of the data collection (no details needed)

Initial Results: Encoder Model



Cletus, describe the results of the encoder model study. Tables and/or graphs with a few bullets should suffice

Initial Results: Decoder Model



Yaman, describe the results of the decoder model study. Tables and/or graphs with a few bullets should suffice

Conclusion: Recommended Two Models



Yaman and Cletus, describe the recommended combination of models

Conclusion: Future Research



Yaman, Cletus, and Sibonginkosi: please use this slide to identify what you have or will do to further improve this research (e.g., new data collection, PERMANOVA, comparing single model design to two model design)

Questions and Comments



Any questions or comments?